

WASP-35b

R-1743

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_140115 · created 2026-08-02T18:50:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456672.80635 +/- 0.0002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.012
Transit depth (Rp/Rs)^2	1.49 +/- 0.28 [%]
Orbital Inclination (inc)	89.97 +/- 1.47
Airmass coefficient 1 (a1)	1.329 +/- 0.016
Airmass coefficient 2 (a2)	-0.1066 +/- 0.0016
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	None
Optimal Aperture	4.76
Optimal Annulus	11.97
Transit Duration (day)	0.1332 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.122 $\pm$ 0.012 vs 0.1241 (deviation 0.17 σ)
Duration fit vs published	0.1332 d vs 0.1317 d (deviation 0.39 σ)
Mid-time O–C	0.5 min
Depth SNR	10.0
Residual scatter	1.15 %

Light curve

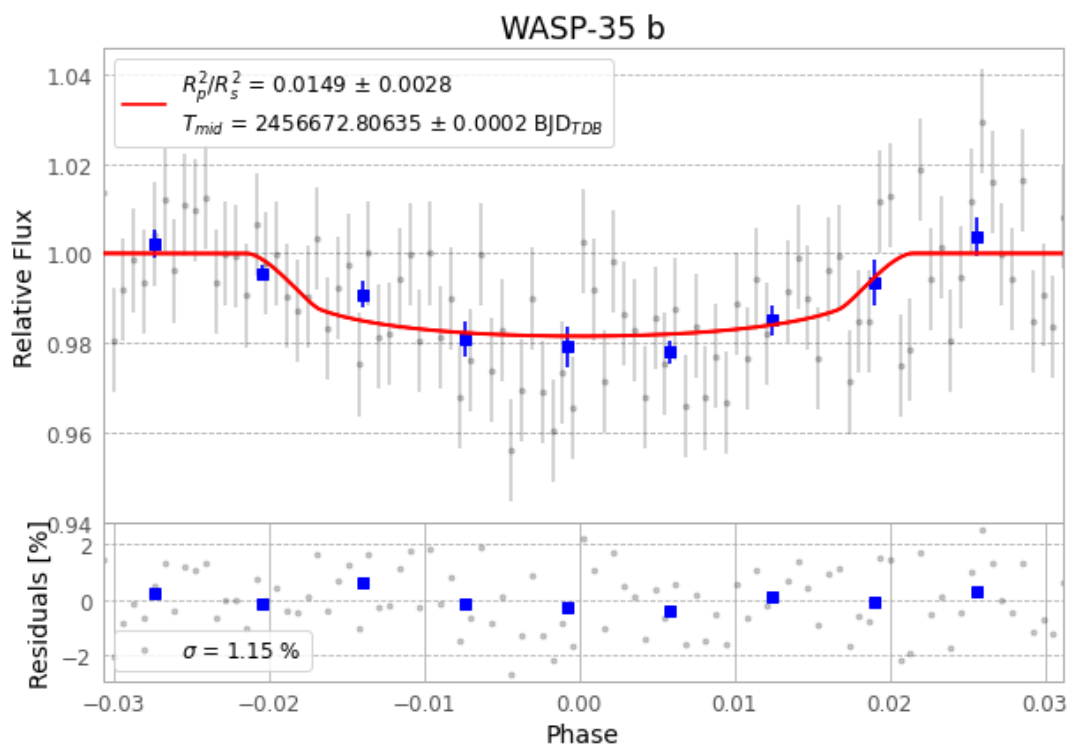


Figure 1 — FinalLightCurve_WASP-35 b_2014-01-14.png (SHA-256 6c2f189be5cbede4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 86fdf157397dc4ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

95 FITS frames (62 MB), WASP-35140115045412.FITS → WASP-35140115093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-140115.log (c0ad6454ad9e...), FinalLightCurve_WASP-35 b_2014-01-14.png (6c2f189be5cb...), FinalLightCurve_WASP-35 b_2014-01-14.csv (c04bbce1d331...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:50Z.